Bri Epstein

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EDUCATION

University of Michigan – Ann Arbor April 2022
Bachelor of Science in Engineering in Computer Science GPA: 3.55/4.00
Minor in Asian Languages and Cultures (Japanese focus)

- Relevant Coursework: Computer Organization, Computer Security, Web Systems, Foundations of Computer Science, Computer Vision, Software Engineering, Computer Game Design and Development, Machine Learning
- Dean's Honor List: Fall 2017, Spring 2018, Fall 2018 and University Honors: Fall 2018, Winter 2020
- Instructional Aide for Computer Game Design and Development: Fall 2021

PROJECTS

Game Development Project "Rideshare Rampage" May 2021 – August 2021

- Developed, with team, game in Unreal Engine in which the player drives customers to destinations around a map to earn money and unlock more areas
- Implemented AI for traffic and pedestrians to continue along paths, and to recover and respond appropriately in multiple scenarios and in response to actions taken by the player

Physical Commonsense Reasoning in Machines (add'l. acknowledgement) June 2020 – May 2021

- Generated ground truth data for evaluation of natural language processing models' physical reasoning when provided physically plausible or implausible stories
- Analyzed dataset for measurement of physical state properties and types of conflicts to evaluate and explain model predictions

Improving Vision-and-Language Navigation Metrics and Datasets May 2020 – September 2020

- Collected human performance data as a benchmark to create a new dataset for evaluating various models' performance in navigating an environment when given natural (human) language instructions
- Explored, with research group, potential improved metrics to reward agents for accurate intermediate decisions even when the correct final goal is not necessarily reached
- Assisted with generation and improvement of progressive dataset, which contains shorter instructions and gauges models' abilities to recognize landmarks along a path

Manga Colorization Using Machine Learning March 2020 – April 2020

- Implemented, with partner, Conditional Generative Adversarial Network (cGAN) architecture to color in the black and white pages of a comic
- Utilized screentone removal and image segmentation to clean original black and white images and prepare them for colorization while preserving important features
- Applied post-processing techniques, such as quantization, to improve the quality of output images

Scalable Search Engine November 2019 – December 2019

- Collaborated with team on end-to-end search engine similar in behavior to commercial search engine
- Implemented indexing with map and reduce tasks to allow for parallel data processing and the use of large datasets
- Applied information retrieval concepts, such as the PageRank search algorithm, to display relevant search results on a graphical user interface

SKILLS

- Computer: C/C++, C#, Python, Java, JavaScript, HTML, Git, Unity, Unreal Engine
- Languages: Limited working proficiency in Spanish and Japanese